Keshav Rajput

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Quantitative researcher / quant developer, new grad. MS CS, NYU Tandon; B.Tech CS and Applied Mathematics, IIIT Delhi.
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EDUCATION

New York University, Tandon School of Engineering Sep 2024 - May 2026
Master of Science (MS) in Computer Science GPA 3.67/4.0

Indraprastha Institute of Information Technology (IIIT) Delhi 2020 - 2024
Bachelor of Technology (B.Tech) in Computer Science and Applied Mathematics, minor in Economics CGPA 8.19/10

Coursework: Stochastic Processes, Statistical Inference, Convex Optimization, Game Theory, Econometrics, Valuation and Portfolio Management, Machine Learning

PROJECTS

trading-agent (not public) | Python, pandas, LightGBM, TabPFN, Chronos-2 2026
- Built a cost-aware backtester: purged walk-forward with a 5-day embargo, CPCV, Deflated Sharpe, probability of backtest overfitting (0.06) and a random-portfolio null; 14 models on 8 US ETFs, 2011 to 2024.
- No candidate beat volatility-targeted equal weight after costs (best Sharpe 0.68 vs 0.81), so none was promoted; the zero-shot Kronos model called 5-day direction right only 46.6% of the time.
- Wrote a paper-trading runner with risk controls: 10% target volatility, 35% per-ETF cap, exposure halved at a 5% drawdown, kill switch at 10% drawdown or a 3% daily loss; four gates before real money.

Non-stationary bandits | Python, NumPy, Matplotlib 2026
github.com/happyc0der/multi_armed_bandit_algorithms
- Implemented five piecewise-stationary bandit algorithms (TS-GE, AdSwitch, M-UCB, UCB1, epsilon-greedy) behind one interface and benchmarked them from 2 to 128 arms.
- Audited TS-GE against its paper and found its Beta update reversed; fixing it cut TS-phase regret from 21,860 to 174, and showed mandatory probing causes 75.9% of its total regret.
- Made AdSwitch 59x faster (218.6 s to 3.7 s at K=16, T=1e5) and cut its memory from 2,048 MB to 272 MB at K=128, T=1e6, with bit-identical output.

Econometrics re-analysis (team of 5) | R, fixest, data.table, Python, pandas 2026
github.com/utkar22/Econometric-Analysis-of-Agriculture-and-Health-Across-India
2023 coursework with four teammates in Utkarsh Arora's repo; the 2026 re-analysis, pipeline, models, review and CI are mine.
- Re-analyzed a 2023 econometrics team project on 673 Indian districts: reproduced both regression tables to 6 decimals, then showed the crop indices were artifacts of row order.
- Built a Python ETL for a 3,585-row district-year panel from Census 2011, NFHS-4 and IMD data; R (fixest) panel regressions with fixed effects and state-clustered errors show the headline effects vanish (within-R2 below 0.02).
- Wrote the review, fixed 7 run-breaking bugs in the archived scripts, and added CI that reruns the deterministic pipeline and fails on any changed table.

Kaggle S6E9, EV purchases | Python, LightGBM 2026
github.com/happyc0der/kaggle-s6e9-ev-purchases
- Finished 287th of 3,575 teams (top 9%) in Kaggle Playground S6E9 (ROC AUC): private 0.94545 against a winning 0.94602, out-of-fold 0.94648.
- Engineered features with nested target encodings over the synthetic generator's artifacts on 668,665 rows with 20-fold cross-validation; blended LightGBM, CatBoost and a 15-seed MLP in logit space.
- Judged 192 experiments by paired DeLong hypothesis tests; label resampling showed a perfect model would score 0.9459 +/- 0.0003, so the remaining gap is label noise.

COMPETITIONS

- **Biohub Cell Tracking, Kaggle**, 2026. Final rank 591 of 3,947 teams (top 15%); private leaderboard 0.918, public 0.952.
- **TrafficFlowBench**, 2026. Public leaderboard 0.889 against a 0.553 baseline (LightGBM queue forecasts, a PyTorch U-Net, an NNLS origin-destination solver). In progress, ends Nov 7, 2026.

SKILLS

Languages: Python, C, C++, Kotlin, TypeScript, JavaScript, SQL, Java, R, Bash, eZ80 assembly, Monkey C

Quant: backtesting (purged walk-forward, CPCV, Deflated Sharpe, PBO), financial modeling, econometrics (fixest, fixed effects, clustered SEs), time-series forecasting, Kalman filters, bandits, Markov chains

ML: PyTorch, deep learning, scikit-learn, LightGBM, XGBoost, CatBoost, Hugging Face transformers, NumPy, pandas, polars

Data: PySpark, Databricks, SQL, PostgreSQL, SQLite, pandas, Parquet, time-series forecasting, statistics (hypothesis testing, regression), econometrics (fixest), Streamlit, Altair, Matplotlib

LEADERSHIP AND ACTIVITIES

- Events head, Finnexia (finance club), IIIT Delhi; organized "The Critical Bid"